

Akshya Maharjan, Om Mahato, Matthew Haling Htang, Edger Becerra (Group 11)

CSCE 2130 – 001

Amir Mirzaeinia

October 4, 2026

Probability and Statistical Modeling of Web Traffic

Introduction

In this project, we used the WorldCup98 Web Server Logs Dataset to study web traffic. We looked at different parts of the data, such as the number of requests per minute, request types, the time between requests, and the size of web objects.

We also used different probability models to see how well they could describe the web traffic. Some of the models we looked at were the Poisson, Binomial, Exponential, and Normal distributions. We also compared exact probability calculations with Normal approximations.

The main goal of this project was to understand how web traffic behaves and see how probability and statistics can be used to describe it.

Web Traffic Behavior and Variability

One thing we noticed from the dataset is that the number of requests was not the same every moment. Some moments had more requests, while other moments had fewer requests. This shows that web traffic changes over time. The average number of requests per minute was about 2112.31 requests. We used this value as the average rate, or λ , for the Poisson calculations.

The traffic also has a lot of variation. Real web traffic does not always happen at the same rate. For example, there can be times when many requests happen close together and other times when there are fewer requests. Because of this, the actual data may not perfectly match a theoretical probability distribution. The probability models are useful for describing the general behavior, but they do not necessarily explain every request in the dataset.

Probability Models

1. Poisson Model

We used the Poisson distribution to study the number of requests that happen during a fixed amount of time.

The average number of requests was:

$\lambda = 2112.31$ requests per minute

The Poisson distribution is useful for counting how many events happen during a certain period.

We also compared the exact Poisson probability with a normal approximation. Since λ was very large, the Normal approximation was reasonable for this calculation. For the calculation, we used **$k = 2205$** . We calculated the exact Poisson probability first and then calculated the Normal approximation using a continuity correction.

The purpose of this comparison was to see how close the Normal approximation was to the exact Poisson result.

2. Binomial Model

We also looked at a binary event based on whether a request was Type 1 or not.

The probability of a Type 1 request was approximately:

$P(\text{Type 1}) = 0.8545$

This means that about **85.45%** of the requests in the dataset were Type 1 requests.

For the Binomial calculation, we used **$n = 1000$** requests and **$k = 870$** as the selected value.

We calculated the probability using the exact Binomial distribution and then used a normal approximation with a continuity correction. Since the number of trials was large, the Normal approximation was useful for comparing the result with the exact Binomial calculation.

3. Interarrival Time

Interarrival time is the time between two requests. We looked at this because it helps us understand how often requests arrive. An Exponential distribution can be used to model interarrival times when requests are happening independently at a constant rate.

However, real web traffic does not always work this way. Requests can happen in groups or bursts, and the traffic rate can change over time. Because of this, the actual interarrival times may not perfectly follow an Exponential distribution.

4. Object Size

We also looked at the size of the web objects being requested.

Object sizes can be very different from one request to another. Some objects can be small, while others can be much larger. Because of this, the distribution of object sizes can be uneven or skewed.

A Lognormal distribution can be useful for data that is positive and has this type of shape. We compared the model with the actual object-size data to see how well it represented the data. The results from the earlier analysis should be used when explaining exactly how well the model fit the data.

Normal Approximations

We used the Normal distribution to approximate both the Binomial and Poisson distributions.

For the Binomial distribution, the mean was calculated using:

$$\mu = np$$

and the standard deviation was:

$$\sigma = \sqrt{np(1 - p)}$$

We also used a continuity correction when calculating the Normal approximation.

For the Poisson distribution, the mean and variance are both equal to λ . Therefore, we used:

$$\mu = \lambda$$

and

$$\sigma = \sqrt{\lambda}$$

We again used a continuity correction.

Because the expected number of events was large, the Normal approximation was a good way to compare with the exact Binomial and Poisson calculations.

Why the Models May Not Match the Data Exactly

The probability models are based on certain assumptions. Real web traffic does not always follow all these assumptions. One reason for differences is that web requests may not be completely independent. For example, when a user loads a webpage, several requests can happen close together. This can create groups or bursts of requests.

Another reason is that the traffic rate can change over time. Some periods can have more activity than others. A Poisson model assumes a certain average rate, so changes in the actual rate can cause differences between the model and the data. The Binomial model also assumes that each trial is independent and has the same probability of success. Real web requests may not always behave this way.

The same idea applies to the Exponential model. It works best when requests arrive independently and at a constant rate. If the traffic changes or requests happen in groups, the actual interarrival times can be different from the model. Because of these reasons, it is normal for the actual data and theoretical models to have some differences.

What we can Conclude from the Data

Some of our results come directly from the dataset.

For example, the average number of requests was about 2112.31 requests per minute. We also calculated that the probability of a Type 1 request was about 0.8545. The variation in the number of requests, interarrival times, and object sizes can also be seen directly from the data and the graphs in the notebook.

However, when we say that a certain probability model is a good model for the data, we are making some assumptions. For example, the Poisson model assumes a certain type of request behavior, while the Binomial model assumes independent trials with the same probability.

Therefore, the model results should not be treated as an exact description of the web traffic. They are mathematical models that help us understand the general behavior of the data.

Summary Table

Variable/Model	What we looked at	Main Result
Requests per minute	Number of requests during each minute.	Average was about 2112.31 requests/minute.
Poisson model	Number of requests during a fixed time.	Used to model request counts.
Binary request event	Whether a request was Type 1	$P(\text{Type 1}) \approx 0.8545$
Binomial model	Number of Type 1 requests out of 1000	Compared exact Binomial and Normal approximation

Interarrival time	Time between two requests.	Used to study how requests arrive.
Exponential model	Distribution of interarrival times.	Compared with the actual request data.
Normal model	Approximation of large Binomial/Poisson values.	Used with continuity correction.
Object size	Size of requested objects.	Used to study the variation in object sizes.
Lognormal model	Model for positive, skewed data.	Compared with the object-size data.

Conclusion

Overall, this project helped us understand how web traffic changes over time. The number of requests was not the same in every time-period, and some requests happened very close together. The Poisson model was useful for studying the number of requests in a fixed time-period, while the Binomial model was useful for studying a binary event such as a Type 1 request. We also used the Exponential model for interarrival times and considered the Lognormal model for object sizes.

The Normal approximation was useful when the number of events was large because it gave us a simpler way to estimate probabilities. By comparing the exact Binomial and Poisson results with their normal approximations, we could see how close the results were. Overall, the models helped us understand the web traffic data, but the actual traffic does not always follow the assumptions of theoretical models. This shows that real-world web traffic can be more variable and complicated than a simple probability model.

Graphs

The empirical probability mass function (PMF) of X .
The empirical CDF of X and use it to estimate probabilities such as $P(X \leq x)$ and $P(a < X \leq b)$.

